LESSON 6 • TERM 1 • LEVEL 1

# Tens Jump, Ones Hop

Adding and taking away tens and ones

Hi, I'm Mi. Let's notice, test and explain why!




TODAY WE WILL

Jump in tens, hop in ones, and say which digit changes.

2–8 MIN

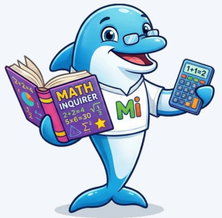
# Warm-up: count in tens

**Clap on every number**

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **10** | 20 | **30** | 40 | **50** | 60 | **70** | 80 | **90** | 100 |

Now start at 3 and add ten each time.

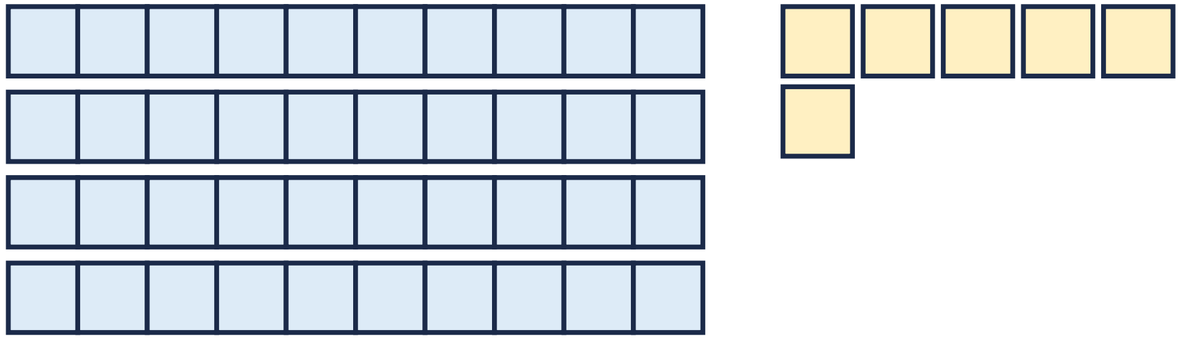
| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **3** | 13 | **23** | 33 | **43** | 53 | **63** | 73 | **83** | 93 |




Let's count along!

# Tens and ones

2–8 MIN



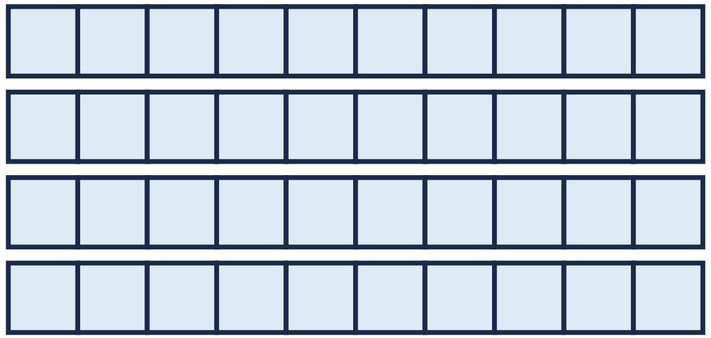
**4 tens and 6 ones**

A bar is a ten. A square is a one. How many tens? How many ones?

**52** | 70 | **83**

2–8 MIN

# Ten and some more



4 tens

40 and 5 more is 45. Count on from the tens.

30 + 4 | 50 + 7 | 80 + 2

8–12 MIN

# Sneak-peek puzzles

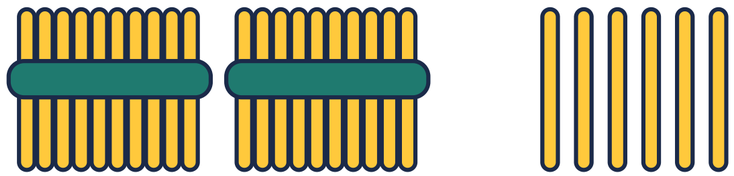
Ideas we have not learnt yet. A guess is a great start. Nobody is marked.

PUZZLE 1: anyone can try

Mei has these sticks.

Then she gets another bundle of 10 sticks.

How many sticks does Mei have now?



PUZZLE 2: think

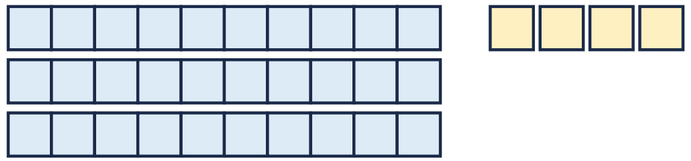
Ravi says 43 + 20 = 45.

Is Ravi right? Write Yes or No. Then write how you know.

12–28 MIN

# Explore: jump with bars

1. Build 34: 3 bars and 4 squares.
2. A big jump is a bar. Add one bar. Say the number.
3. Jump again. Say the number.
4. Jump back: take a bar away. Say the number.

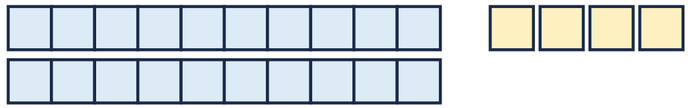


34

12–28 MIN

# Explore: hop with squares

1. Build 24: 2 bars and 4 squares.
2. A small hop is a square. Hop forward 3 times.
3. Hop back 2.
4. Which digit changed?



24

12–28 MIN

# Explore: jump and hop

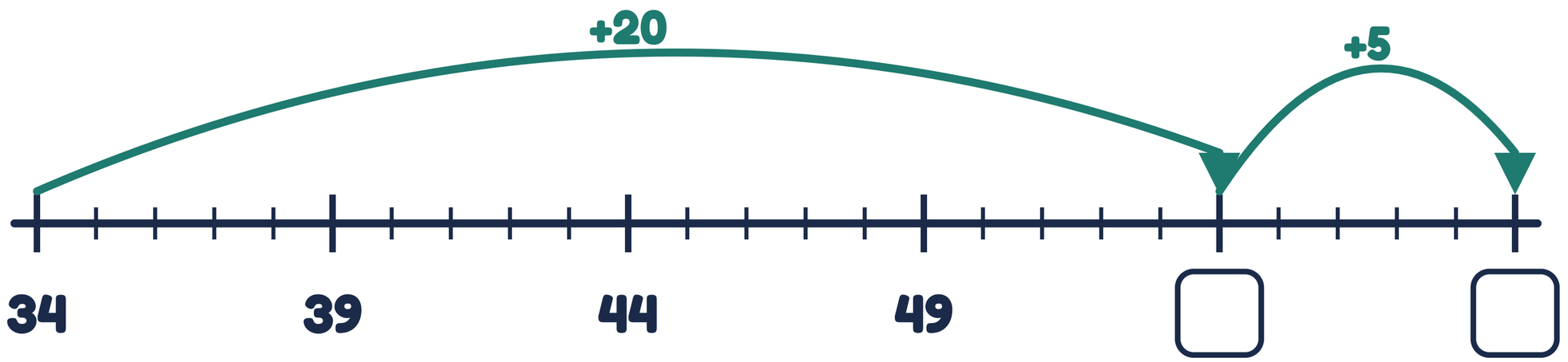
Build the first number. Jump the tens. Then hop the ones. Show a thumbs up when you have built it.

**34 + 25** | **41 + 38** | **68 - 24**

12–28 MIN

# Draw the jumps and hops

1. Draw a dot. Write the start number.
2. Draw a big arc for the jump. Write +20.
3. Write where you land.
4. Draw a small arc for the hop. Write +5. Write where you land.



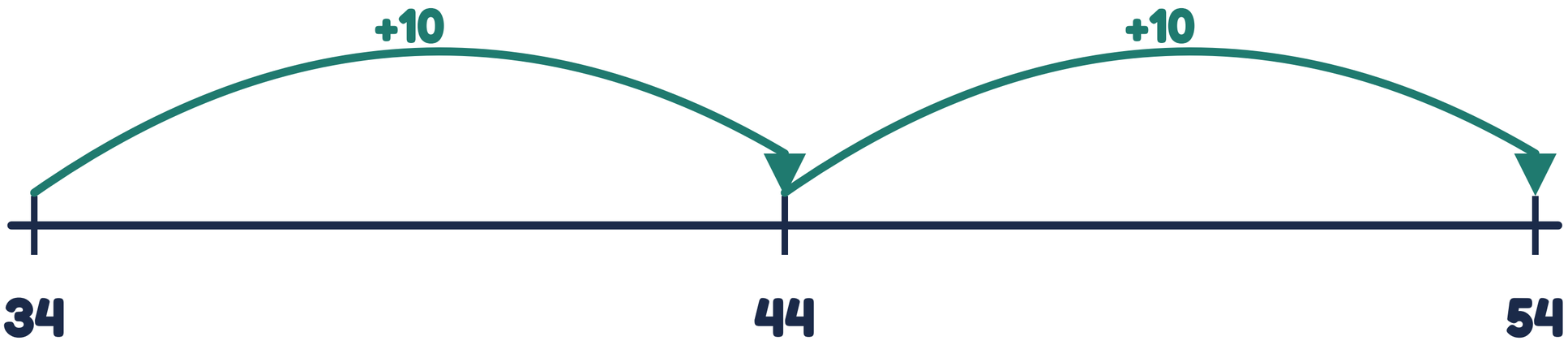
12–28 MIN

# What did we notice?

- Which one changes the tens digit, a jump or a hop?
- Which one changes the ones digit?
- Does it matter if you hop first or jump first?
- How many ones can we hop before we reach ten?

28–38 MIN

# Jump forward: add tens

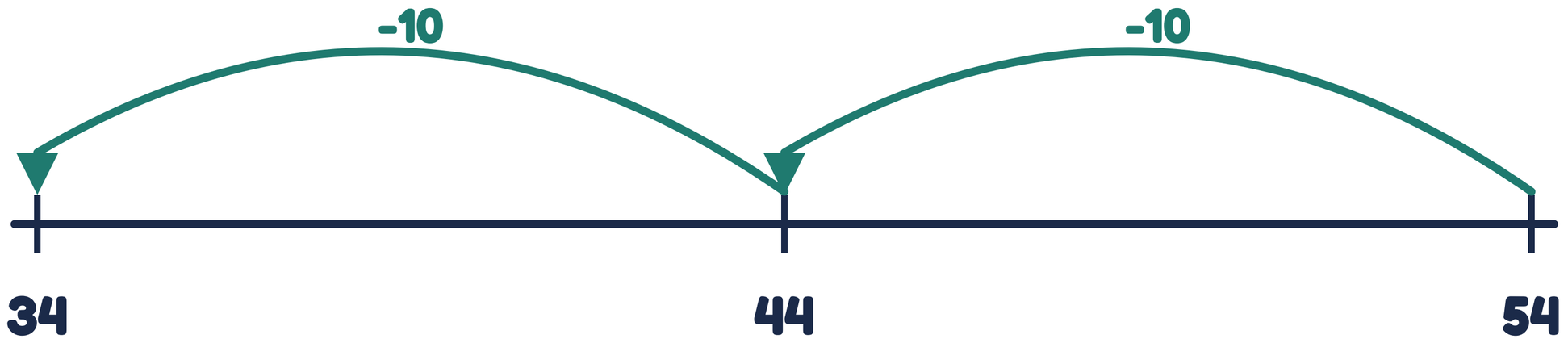



Each jump adds 1 ten. The ones digit stays the same.

**34 + 20 = 54**

28–38 MIN

# Jump back: take away tens

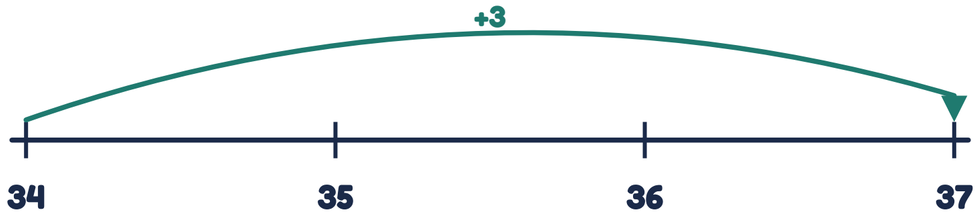



Each jump back takes away 1 ten. The ones digit stays the same.

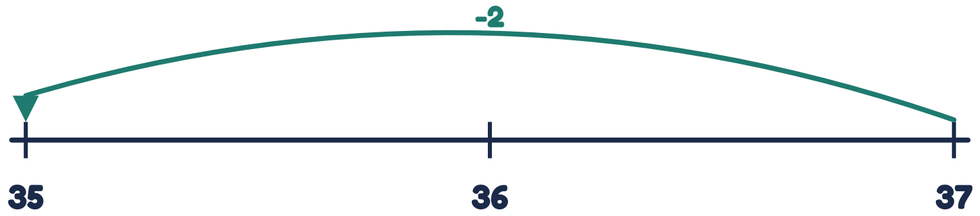
**54 - 20 = 34**

28–38 MIN

# Hop: ones




34 + 3 = 37

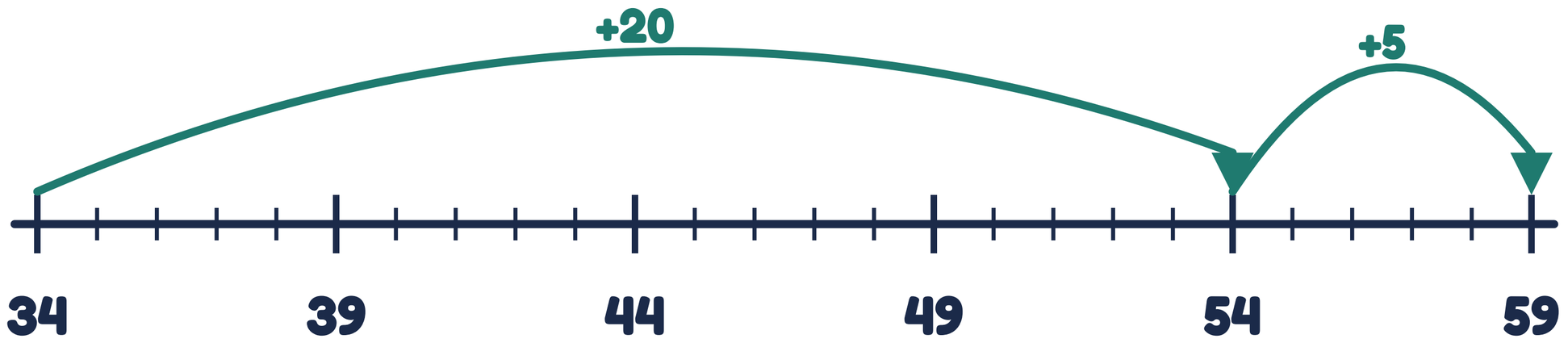



37 - 2 = 35

**A hop changes the ones digit. The tens digit stays.**

28–38 MIN

# Jump, then hop: add

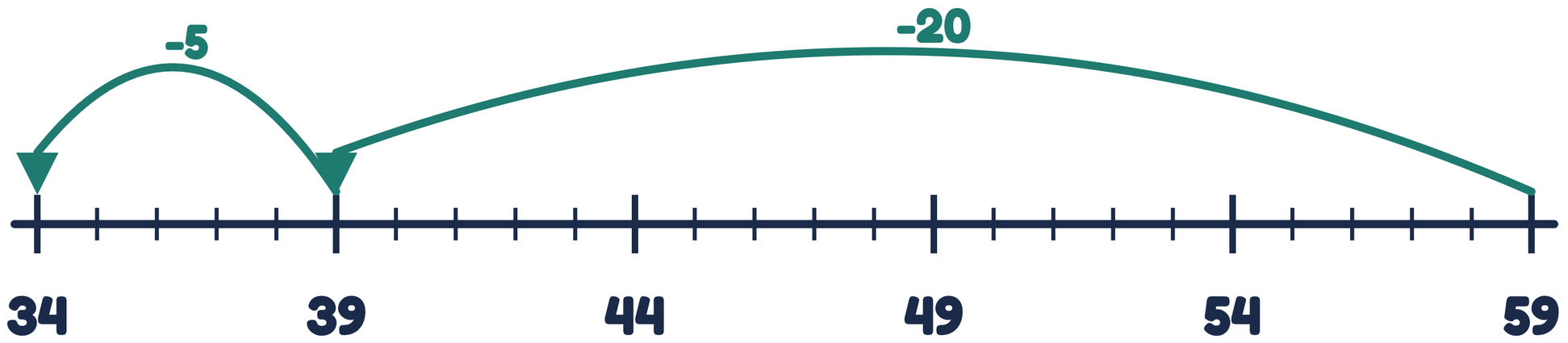



25 is 2 tens and 5 ones. Jump the tens first. Then hop the ones.

**34 + 25 = 59**

28–38 MIN

# Jump, then hop: take away




25 is 2 tens and 5 ones. Jump back the tens first. Then hop back the ones.

**59 - 25 = 34**

28–38 MIN

# Your turn: write the answer

Write on your paper. About 2 minutes

45 + 30

78 − 20

64 − 3

52 + 36

28–38 MIN

# Check: write the answer

45 + 30 = 75

78 - 20 = 58

64 - 3 = 61

52 + 36 = 88

28–38 MIN

# Jump or hop?

| | | | |
|---|---|---|---|
| 30 | 6 | 50 | 2 |

Tens are a jump. Ones are a hop.

38–42 MIN

# Jump the tens, hop the ones

A big jump for every ten. A small hop for every one.

START ON 34

Jump forward 2 times.

Hop forward 3 times.

Jump back 1 time.

Hop back 4 times.

Say the number each time. Sip some water.

42–70 MIN

# Workbook page 1: Start here

Questions 1 to 3. About 9 minutes

**S1**

**These are Mei’s sticks. She gets another bundle of 10 sticks. Fill in the blanks.**



(a) Now Mei has _____ bundles of ten.

(b) She also has _____ single sticks.

(c) Mei has _____ sticks altogether.

**S2**

**In the box, draw 3 tens as 3 tall bars and 4 single sticks. Next, draw 2 more tall bars. How many sticks are there altogether?**

There are _____ sticks altogether.

**S3**

**Jump forward. Fill in the blanks.**

(a) 23 + 10 = _____

(b) 23 + 40 = _____

(c) 51 + 30 = _____

(d) 7 + 60 = _____

# Check page 1

42–70 MIN

| | |
|---|---|
| S1 | 3 bundles, 7 singles, 37 sticks |
| S2 | 54 (5 bars drawn and 4 singles) |
| S3 | 33, 63, 81, 67 |

# Workbook page 2: Start here

42–70 MIN

Questions 4 to 7. About 9 minutes

**S4**

**Jump or hop. Fill in the blanks.**

(a) 34 - 10 = _____

(b) 58 - 20 = _____

(c) 58 - 3 = _____

(d) 62 + 7 = _____

**S5**

**Ali has 68 pencils. He gives 3 boxes of 10 pencils to his friend. How many pencils does Ali have left?**

There are _____ pencils left.

**S6**

**Ben says 45 + 20 = 47. Fill in a to c.**

(a) Is Ben correct? Circle Yes or No.

(b) The right answer is _____.

(c) Ben added 2 _____ by mistake. He should add 2 _____.

**S7**

**Start at 36. On a number line in the box, draw 2 jumps of ten. Where do you land?**

I land on _____.

# Check page 2

42–70 MIN

| | |
|---|---|
| S4 | 24, 38, 55, 69 |
| S5 | 38 pencils |
| S6 | No. The right answer is 65. He added 2 ones instead of 2 tens. |
| S7 | 56 (two jumps of ten from 36: 46, then 56) |

42–70 MIN

# Choose your next page

| Page | What it is |
| --- | --- |
| **Think harder (page 3)** | Jump and hop in a new setting |
| **Go further (page 4)** | Bigger journeys on the number line |
| **Investigation (page 5)** | Always, sometimes or never? |
| **Check** | Finish and check your own work |

**Everyone may choose. You can swap at any time.**

# Talk and explain

70–76 MIN

MEI SAYS

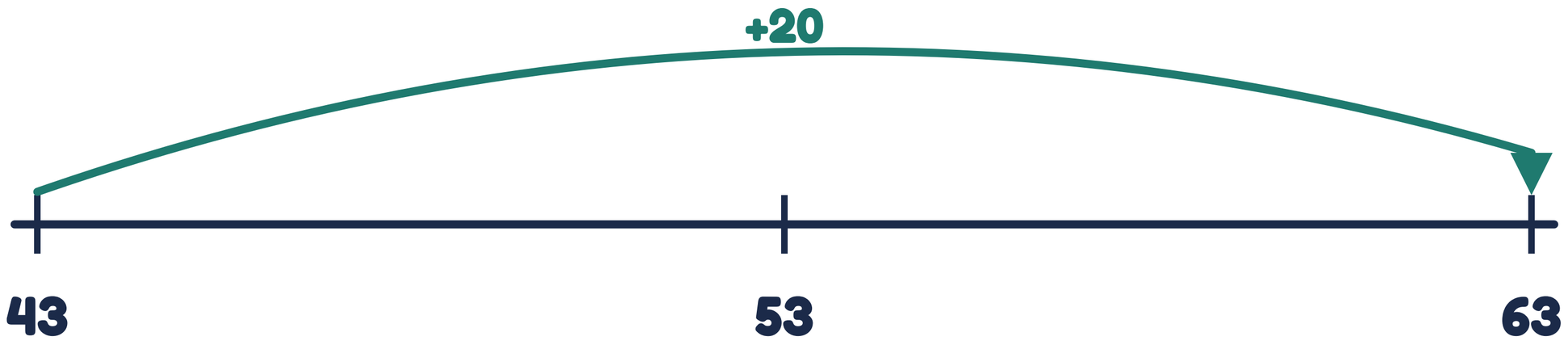
When you jump forward 2 tens, the ones digit changes.

Is Mei right? Use your kit to show why.

I think … because …

70–76 MIN

# Back to the sneak-peek




**No. 43 + 20 = 63. Twenty is 2 tens, so we jump.**

76–82 MIN

# Exit ticket

On your own. No looking back. About 3 minutes

**E1** **Add.**

46 + 30 = _____

**E2** **Subtract.**

(a) 74 - 30 = _____

(b) 74 - 3 = _____

**E3** **Mei is on 52. She jumps forward 2 tens, then hops forward 3. Fill in the blanks.**

(a) After the jump she is on _____.

(b) After the hop she is on _____.

# Check the exit ticket

76–82 MIN

| | |
|---|---|
| E1 | 76 |
| E2 | 44 and 71 |
| E3 | 72 and 75 |

# Look-back

82–85 MIN

**Three minutes. There are no right answers. You can pass. Nobody reads it out.**

| | |
|---|---|
| **Something I did well** | write or draw one thing |
| **The hardest part was …** | ☐ knowing if it is a jump or a hop ☐ taking away tens ☐ jump, then hop<br>☐ the number line ☐ something else |
| **What helped me** | ☐ the bars and squares ☐ drawing the jumps ☐ trying again ☐ I asked for help<br>☐ I was not stuck |
| **Next time I will try** | ☐ drawing it ☐ using objects ☐ checking my answer ☐ asking a question |
| | ☐ I am still stuck (my tutor will come and help) ☐ I would like to pass |

**Take your time. You can pass.**



85–90 MIN

# Well done, inquirers!

Show a grown-up one thing you found out today.

NEXT LESSON

Take Away, Count Up. Bring your kit, a pencil, a pencil of another colour, paper and an envelope.

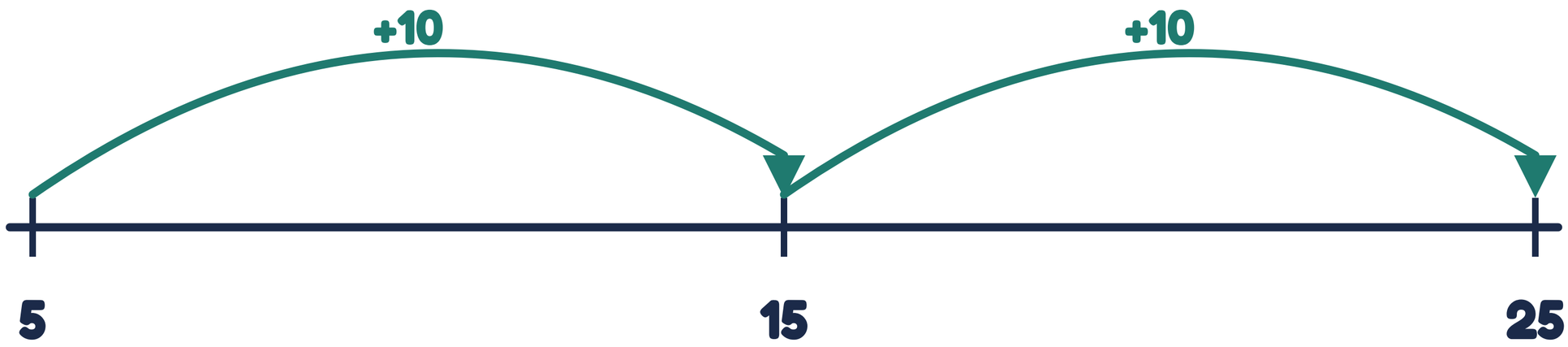


OPTIONAL

# Go further: what if?

Start at 5. Jump in tens. What is the last number you land on before 100?

How many jumps is that?



OPTIONAL

# If we have time: pick one

| Activity | What we do |
| --- | --- |
| **Is it always true?** | Add a ten to 34, 67 and 90. What changes? |
| **Make a puzzle** | Write a jump and hop story for your grown-up |
| **Teach it back** | Show how to work out 52 + 31 with a jump and a hop |

**Optional. Everyone may join.**